Alejandro Medina

Waco, TX | 512-412-0965 | alejmedinajr@gmail.com | LinkedIn | GitHub

EDUCATION

Ph.D. in Computer Science
Baylor University, Waco, TX **August 2024 – Present**

- Research Interests: Quality Diversity and Evolutionary Computation; Reinforcement Learning; Generative Models; Machine Learning for Bioinformatics and Behavioral Data; Optimization Algorithms.

Bachelor of Science in Computer Science
Southwestern University, Georgetown, TX **August 2021– May 2024**

- Distinctions: Magna Cum Laude (GPA 3.88), Paideia Scholar, Dean's List (August 2021– May2024)
- Minors: Mathematics and Data Science
- Honors: Pi Mu Epsilon (President), Upsilon Pi Epsilon (Vice President), Omicron Delta Kappa
- Organizations: Computer Science Club, Latinos Unidos (Event Coordinator), Honor Code Council (Vice President), LatinXcel, Mathematics Club, University Committee on Discipline

PUBLICATIONS (* = accepted papers not yet formally published)

- **A. Medina** and M. L. Benton
  Motif Diversity in Human Liver ChIP-seq Data Using MAP-Elites
  *Genetic and Evolutionary Computation Conference (GECCO Companion 2026)**

- **A. Medina**, M. Richey, M. Mueller, and J. Schrum
  Evolving Flying Machines in Minecraft Using Quality Diversity
  Proceedings of the Genetic and Evolutionary Computation Conference (GECCO 2023), ACM, pp. 1418–1426.

- B. Anthony, **A. Medina**, and M. Mueller
  Prioritizing Self, Team, or Job: Trends in Sincerity in Cooperative Polls
  Proceedings of the International Conference on Cooperative Design, Visualization, and Engineering (CDVE 2022), LNCS 13492, Springer, pp. 34-44.

WORK EXPERIENCE

Graduate TA | **Baylor University**, Waco, TX **August 2024 – Present**

- Supported instruction across courses: Software Project Management, Computer Science I & II.

Undergraduate Research Mentor | **Southwestern University**, Georgetown, TX **January 2024 – May 2024**

- Mentored first-year students in a small cohort on a research project related to voter theory algorithms.

Business Intelligence and Data Analytics Intern | **Hy Cite Enterprises**, Madison, WI **June 2023 – August 2023**

- Developed machine learning models for predictive analytics, anomaly detection, and clustering, supporting data-driven financial decision-making across business units.
- Identified and resolved performance bottlenecks in the company's data warehouse by analyzing query execution plans and optimizing data pipelines.
- Designed and automated reporting workflows using Python and Snowflake, improving efficiency and reducing manual reporting overhead.
- Independently analyzed large-scale business data and presented actionable insights to non-technical stakeholders, translating technical findings into strategic recommendations.

Computer Science Tutor | **Southwestern University**, Georgetown, TX **January 2023 – May 2023**

- Supported instruction across courses: Explorations in Computing, Computer Science I & II.

SELECTED RESEARCH PROJECTS

Quality Diversity Exploration of DNA Motif Discovery | *MAP-Elites, Bioinformatics, Evolutionary Computation [2026]*

- Companion paper accepted to GECCO Companion 2026; selected for national presentation at:
  - CRA Graduate Student Workshop (Lightning Talk)
  - CMD-IT/ACM Richard Tapia Conference (Graduate Poster Competition).
- Formulated DNA motif discovery as a quality diversity (QD) problem and developed a MAP-Elites framework for evolving position weight matrices (PWMs) under likelihood-based fitness.
- Recovered motifs with fitness comparable to MEME while revealing structural diversity missed by single-solution methods.
- Established QD as an exploratory paradigm for biological sequence analysis.

Alcohol Consumption Phenotyping in Rhesus Macaques (MATRR) | *Machine Learning, Bioinformatics [2025]*

- Reassessed alcohol-intake trajectory classifications using multiple machine learning models on MATRR data.
- Identified instability and limited predictive validity in widely used phenotyping categories, challenging assumptions in foundational studies.
- Demonstrated that standard categorization approaches may obscure meaningful behavioral patterns in longitudinal alcohol consumption data.
- Presented findings at the 2025 AI for Biology & Medicine Symposium.

Quality Diversity for Creative AI in Minecraft | *Evolutionary Computation, MAP-Elites, Generative Systems [2023]*

- Co-authored GECCO 2023 paper introducing MAP-Elites for procedural generation of functional, physics-constrained flying machines in Minecraft.
- Designed novel behavior descriptors and archive structures for generating diverse, high-performing mechanical designs.
- Demonstrated one of the first evolutionary approaches for generating complex, functional flying machines.
- Bridged evolutionary search and creative design in open-ended environments.
- Results presented in poster competitions earning 3rd Place and Honorable Mention awards.

Algorithmic Scheduling and Dial-a-Ride Optimization | *Algorithms, Scheduling, Optimization [2023]*

- Investigated single-vehicle Dial-a-Ride problems under deadline-constrained scheduling settings.
- Implemented and evaluated Earliest Deadline First (EDF) against alternative scheduling strategies.
- Analyzed trade-offs between scheduling policies under constrained routing scenarios.
- Presented results at the CMD-IT/ACM Richard Tapia Conference (2023).

RESEARCH APPOINTMENTS

Graduate Research Assistant | **Baylor University**, Waco, TX **May 2025 – Present**
Advisor: Dr. Mary Lauren Benton

- Lead research on quality diversity approaches for DNA motif discovery, resulting in a GECCO Companion 2026 accepted paper and multiple national presentations (CRA workshop, CMD-IT/ACM Tapia Conference).
- Conducted machine learning analysis of alcohol consumption phenotypes in rhesus macaques (MATRR), evaluating stability and predictive validity of behavioral classifications.

Undergraduate Research Assistant | **Southwestern University**, Georgetown, TX **January 2022 – August 2023**
Advisor: Dr. Barbara Anthony

- Investigated algorithm design and scheduling problems, including Dial-a-Ride optimization under deadline constraints.
- Conducted research on approval voting theory and cooperative polling, contributing to a peer-reviewed conference publication (CDVE 2022).

Advisor: Dr. Jacob Schrum

- Developed quality diversity algorithms for creative AI, contributing to a GECCO 2023 publication on evolving functional flying machines in Minecraft.

PRESENTATIONS *(* = accepted presentations not yet delivered)*

Graduate Presentations

- "Motif Diversity in Human Liver ChIP-seq Data Using MAP-Elites."
  - Poster, GECCO Companion, 2026.*
  - Poster, CMD-IT/ACM Richard Tapia Conference, 2026 (Graduate Poster Competition).*
  - Lightning Talk, CRA Graduate Student Workshop, 2026.

- "Revisiting Predictive Models of Alcohol Consumption in Nonhuman Primates."
  Poster, AI for Biology & Medicine Symposium, 2025.

Undergraduate Presentations

- "SNITCH: Southwestern's Newest Innovation to Cultivate Honor."
  Poster, South Central CCSC Regional Meeting, 2024.
  (3rd Place: Undergraduate Poster Competition)

- "Evaluating an Earliest Deadline First Algorithm for a Dial-a-Ride Problem."
  Poster, CMD-IT/ACM Richard Tapia Conference, 2023.

- "Using Quality Diversity to Evolve Flying Machines in Minecraft."
  Poster, South Central CCSC Regional Meeting, 2023.
  (3rd Place: Undergraduate Poster Competition)

- "Interactive Evolution of Novel Shapes in Minecraft."
  Poster, South Central CCSC Regional Meeting, 2023.
  (Honorable Mention: Undergraduate Poster Competition)

- "Prioritizing Self, Team, or Job: Trends in Sincerity in Cooperative Polls."
  Oral Presentation, 19th International Conference on Cooperative Design, Visualization and Engineering, 2022.

RESEARCH AWARDS/HONORS

Computing Research Association's Outstanding Undergraduate Researcher Award **(2024)**

- National award recognizing top undergraduate researchers in North America.

CCSC Undergraduate Poster Competition

- *3rd Place (2023, 2024)*
- *Honorable Mention (2023)*

OTHER AWARDS/HONORS

Judge for Research & Internship Day Undergraduate Poster Competition *(2025)*

- Selected to judge research poster competition for Baylor University's Research & Internship Day.

Five Points of Pirate Pride Award *(2024)*

- Award for most exemplary graduating student across all of Natural Sciences.

Grogan Lord Award in Computer Science (March 2024)

- Award for most exemplary graduating student within the Computer Science department.

Designated Matriculation Speaker (2023)

- Selected to give matriculation speech to the incoming freshman class of 2027 at Southwestern University.

2023 Hispanic Scholarship Fund (HSF) Scholar (2023)

- Designated Hispanic scholar from a highly competitive pool of over 124,400 applicants.

Jack Frost Citizen of the Month (2021)

- Recognized for my actions and leadership within my community.

SKILLS & INTERESTS

Technical Domains: Machine Learning, Evolutionary Computation, Optimization, Data Analysis, Bioinformatics
Languages: Python, C++, Java, SQL, R, Rust
Tools: Git, Docker, Snowflake, React, LaTeX

Professional Associations: ACM, IEEE, SACNAS, SHPE, LatinX in AI (LXAI), Computing Research Association (CRA), ALPFA

Languages: English, Spanish

Interests: Soccer, Weightlifting, Running, Violin